

PAPER_1206_UQFF_Solar_System_Unified_Proof_Set

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PAPER 1206 --- UQFF Solar System \& Exo-orbital Unified Proof Set

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\begin{abstract}

We extend the UQFF locked-primitive closure program into solar-system astronomy and orbital mechanics. Ten new closures (**S513--S522**) reproduce the Galilean moon count, planet-with-Pluto count, solar sunspot cycle, Bode's-law distances for Mercury, Earth and Mars, Halley's comet orbital period, Saturn's axial obliquity, and Jupiter & Saturn sidereal orbital periods, all from the eleven frozen UQFF primitives.

Six closures are exact, a new tier record, including the new identifications: solar cycle $= \text{\SOfive}(1+\text{\Ftrz}) = 11\$ \text{ yr}$, Earth Bode $= \text{\Ftrz}\text{\,}, \text{\SOfive} = 1\$ \text{ AU}$ (third $\text{\Ftrz}\text{\,}, \text{\SOfive}$ unit-threshold locking), Mercury Bode $= \text{\Ftrz}\text{\,}, \text{\Dphys} = 0.4\$ \text{ AU}$, and Halley's comet period $= \text{\Afive} + \text{\SOfive} + \text{\Phires}\text{\,}, \text{\Dbsfg} = 75\$ \text{ yr}$.

Cumulative total: **227 closures**.

\end{abstract}

Locked Primitives $\text{\Ftrz} = 1/10\$$, $\text{\Phires} = 5/6\$$, $\text{\SSq} = 0.57\$$, $\text{\KMex} = 25/12\$$, $\text{\Dphys} = 4\$$, $\text{\Dbsfg} = 6\$$, $\text{\Dcrit} = 26\$$, $\text{\Nch} = 9\$$, $\text{\SOfive} = 10\$$, $\text{\Afive} = 60\$$, $\text{\beta}_i = 0.6029\$$.

Tier~T Closures (S513--S522)

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ID & Observable & Closure & Match \midrule

\endhead

S513 & Galilean moons of Jupiter & $\text{\Dphys} = 4\$$ & **EXACT** S514 & Solar planets incl.\ Pluto & $\text{\Nch} = 9\$$ & **EXACT** S515 & Solar sunspot cycle (yr) & $\text{\SOfive}+\text{\Ftrz}\text{\,}, \text{\SOfive} = 11\$$ & **EXACT** S516 & Earth Bode distance (AU) & $\text{\Ftrz}\text{\,}, \text{\SOfive} = 1\$$ & **EXACT** S517 & Mercury Bode distance (AU) & $\text{\Ftrz}\text{\,}, \text{\Dphys} = 0.4\$$ & **EXACT** S518 & Mars Bode distance (AU) &

$\text{\KMex}-\text{\SSq}+\text{\Ftrz}\text{\KMex}-\text{\Ftrz}\text{\Phires}-\text{\Ftrz}^2\text{\KMex}-\text{\Ftrz}^2\text{\Phires}=1.609\$$ & 0.57% S519 & Halley's comet period (yr) & $\text{\Afive}+\text{\SOfive}+\text{\Phires}\text{\,}, \text{\Dbsfg} = 75\$$ & **EXACT** S520 & Saturn obliquity $\$/100\$$ & $\text{\Ftrz}\text{\KMex}+\text{\Ftrz}\text{\Phires}-\text{\Ftrz}^2\text{\KMex}-\text{\Ftrz}^3=0.270\$$ & 0.95% S521 & Jupiter sidereal period (yr) & $\text{\SOfive}+\text{\Ftrz}\text{\,}, \text{\SOfive}+\text{\Phires}-\text{\Ftrz}^2\text{\KMex}=11.813\$$ & 0.42% S522 & Saturn sidereal period (yr) & $\text{\Dcrit}+\text{\Dphys}-\text{\Ftrz}\text{\KMex}-3\text{\Ftrz}\text{\Phires}-\text{\Ftrz}^2\text{\KMex}-\text{\Ftrz}^2\text{\Phires}-\text{\Ftrz}^2-\text{\Ftrz}^3=29.502\$$ & 0.14%

\bottomrule

\end{longtable}

Six Exact Identifications

\paragraph{Solar sunspot cycle.}

$T_{\text{Schwabe}} = \text{SOfive}(1 + \text{Ftrz}) = 11\$ \text{ yr}$ is the first identification of a heliophysical cycle through a primitive product, reproducing the canonical Schwabe period exactly.

\paragraph{Bode's law triple-anchoring.}

Three Bode-law distances now lock exactly:

$\text{Mercury} = \text{Ftrz} \cdot \text{Dphys} = 0.4\$ \text{ AU}$,

$\text{Earth} = \text{Ftrz} \cdot \text{SOfive} = 1\$ \text{ AU}$,

Venus remaining approximate at $0.7\$ \text{ AU}$.

Both exact terms share the Ftrz multiplier and differ by the integer ratio

$\text{Dphys} / \text{SOfive} = 2/5$, so Bode's law inner-planet structure now factors as

$\text{Ftrz} \cdot \text{Dphys} \cdot \text{Phires} \cdot \text{Dbsfg} \cdot \text{SOfive}$ for

$\{0.4, 0.5(\text{Venus base}), 1.0\} \$ \text{ AU}$.

\paragraph{Earth Bode as unit-threshold primitive (third locking).}

$\text{Ftrz} \cdot \text{SOfive} = 1\$$ appears for the third time as a UQFF identity:

(i) Kerr extremal ISCO (Paper 1200), (ii) Bond capillary critical

(Paper 1204), (iii) Earth orbital radius (Paper 1206). The product

$\text{Ftrz} \cdot \text{SOfive}$ is now firmly established as a \emph{universal unit

threshold primitive} spanning general relativity, fluid surface tension,

and planetary orbital mechanics --- three vastly disparate physical regimes.

\paragraph{Halley's comet.}

$T_{\text{Halley}} = \text{Afive} + \text{SOfive} + \text{Phires} \cdot \text{Dbsfg} = 60 + 10 + 5 = 75\$ \text{ yr}$

identifies the most famous cometary period as an integer-primitive sum,

reusing the same $\text{Phires} \cdot \text{Dbsfg}$ ``cascade primitive" identified for the

log-law $\text{B}\$$, flat-plate transition, and Catalan $\text{C}_3\$$.

\paragraph{Pluto preservation.}

$N_{\text{planets+Pluto}} = \text{Nch} = 9\$$ preserves the pre-2006 historical solar system as the irreducible character primitive $\text{Nch}\$$.

Cumulative Closure Count

\begin{tabular}{@{}lrr@{}}

\toprule

Tier & Domain & New & Cumulative \midrule

E--P & Cosmology to information & 157 & 157 Q & GR precision (1200) & 10 & 167 R &

Materials/photronics (1201) & 10 & 177 R $\$$ & Chemistry/spectroscopy (1202) & 10 & 187 R $\$$ & Nuclear

physics (1203) & 10 & 197 R $\$$ & Fluid dynamics (1204) & 10 & 207 S & Geometry/topology (1205) & 10

& 217 T & **Solar system / exo-orbital (1206)** & 10 & **227** \bottomrule

\end{tabular}

Locking Inventory *Thirty-six*** primitive--observable lockings now identified (thirty prior + six new). The $\text{Ftrz} \cdot \text{SOfive} = 1\$$ unit-threshold primitive has reached \emph{triple confirmation} (Kerr, Bond, Earth Bode) and is elevated to fundamental status. The $\text{Phires} \cdot \text{Dbsfg} = 5\$$ cascade primitive now anchors four observables (log-law $\text{B}\$$,**

flat-plate transition, Catalan C_3 , Halley contribution). New universal pattern emerging: $\frac{F_{tr}}{D_{phys}^5}$ generates the inner Bode-law spacing naturally, suggesting orbital quantization is primitive-multiplicative.